LLM LIMITATIONS QUICK-REFERENCE CARD

AI Engineer Ready · Week 2 — What Can Go Wrong, and How to Guard Against It

Golden Rule → LLMs predict plausible tokens, not verified facts — always plan for that

1. How LLMs "Reason"

LLMs do not reason like humans — they perform pattern-based token prediction with internal representations that can **approximate** reasoning behaviors. Practical takeaway: better prompts improve output structure, tools + retrieval improve factual quality, and verification steps reduce hallucinations.

2. Common Limitations & Mitigations

Limitation	What it looks like	Mitigation
Hallucinations	Confident, fluent, but factually wrong answers	RAG (ground answers in retrieved sources), citations
Knowledge cutoff	No awareness of events after training data ended	Retrieval / live tools / web search
Prompt sensitivity	Small wording changes shift the output a lot	Structured prompts, few-shot examples, testing
Bias & safety	Reflects biases present in training data	Evaluation loops, human-in-the-loop review
Context window limits	Long inputs get truncated or "forgotten"	Chunking, summarization, RAG retrieval

3. One-Page Symptom -> Cause -> Fix

Symptom	Likely cause	Fix
Answer sounds right but is wrong	Hallucination — no grounding in real data	Add RAG / require citations / ask model to say "I don't know"
Model doesn't know a recent event	Knowledge cutoff	Feed it current info via retrieval or a tool call
Output changes wildly on rerun	High temperature or prompt sensitivity	Lower temperature; make the prompt more specific/structured
Model "forgets" earlier instructions	Context window exceeded	Shorten input, summarize history, or chunk + retrieve
Output feels one-sided or unfair	Training-data bias	Add review step, diversify prompts/examples, human review

Remember before shipping

Never present raw LLM output as verified fact in a production system without a grounding step (RAG, tool call, or human review) — treat every answer as a draft that needs a check, especially for numbers, dates, names, and citations.

4. Session Wrap-Up — You Should Now Be Able To

Skill
Explain what a language model predicts

Skill
Describe tokens, tokenization, and vocabulary
Work with embeddings and cosine similarity
Understand transformer and attention basics
Compare BERT and GPT at a high level
Identify common LLM limitations and their mitigations